

MediSync AI

Hospital Management System User Manual & Operational Guide

A Comprehensive Guide for Administrators, Healthcare Providers,
Receptionists, and Patients

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1. System Overview & Architecture

MediSync AI is an enterprise-class Hospital Management System designed to modernize clinical and administrative workflows. By integrating clinical decision support, role-based dashboards, and automated queue management, MediSync AI minimizes operational overhead and optimizes patient care delivery.

Core System Stack:

- **Frontend Web Client:** Built with React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons, and Recharts.
- **State & Persistence:** Managed via Zustand with local/remote state synchronization.
- **Backend API Server:** Node.js with Express, TypeScript, and JWT-based authentication.
- **Database Layer:** PostgreSQL in production environments with SQLite fallback support.
- **Containerization:** Full Docker and Docker Compose environment configuration.

2. Getting Started & Installation

Follow the steps below to initialize and deploy the system locally or in containerized environments:

```
# 1. Clone the repository
git clone https://github.com/your-org/medisync-ai.git
cd medisync-ai

# 2. Install and launch Web Frontend
npm install
npm run dev    # Runs at http://localhost:5173

# 3. Install and launch API Server
cd backend && npm install
npm run dev    # Runs at http://localhost:5000

# 4. Alternative Docker Launch
docker-compose up --build -d
```

3. User Roles & Authentication

MediSync AI enforces multi-tenant Role-Based Access Control (RBAC). User permissions and portal views are customized according to authenticated staff roles.

Role	Primary Responsibilities	System Scope
Admin	System config, staff management, financials, analytics	Full System Access
Doctor	Consultations, EHR records, e-Prescriptions, digital signing	Clinical & Patient Portal
Receptionist	Walk-in registration, token issuing, queue management	Front Desk & Queue Module
Patient	Self-booking, viewing prescriptions, invoice payment	Personal Portal
Pharmacy	Prescription verification, stock checks, medication dispensing	Pharmacy Module

Demo Account Credentials:

Role	Email Address	Password
Admin	admin@medisync.com	password123
Doctor	doctor@medisync.com	password123
Receptionist	receptionist@medisync.com	password123
Patient	patient@medisync.com	password123
Pharmacy	pharmacy@medisync.com	password123

4. Admin Portal Guide

The Admin Portal acts as the control center for system configuration, medical staff management, and financial analysis.

- **Executive Dashboard Widgets:** Real-time KPI summaries including total registered patients, active medical personnel, daily appointment volumes, monthly revenue, pending invoices, emergency alerts, and bed

occupancy.

- **Staff & Physician Management:** Onboard new doctors, set department allocations, configure consultation fees, and upload physician digital signature files.
- **Financial & Invoice Management:** Generate billing statements, apply service fee items, track payment states, and process invoice payouts.
- **Analytics Engine:** Interactive charting for department performance, revenue trends, and operational bottleneck detection.

5. Doctor Portal Guide

Designed specifically for clinical practitioners to manage patient appointments, review medical histories, and issue prescriptions.

- **Consultation Desk:** Manage patient flow across statuses: *Scheduled* → *Waiting* → *In Progress* → *Completed*.
- **EHR & Medical History:** Immediate access to historical diagnoses, vital sign trends, past prescriptions, and allergy alerts.
- **e-Prescribing Tool:** Add medication names, dosages (e.g., 500mg), administration frequencies (e.g., 1-0-1), duration, and custom clinical notes. Automatic application of the doctor's verified digital signature.

6. Receptionist Portal Guide

Optimized for front-desk efficiency, walk-in triage, patient registration, and queue coordination.

- **Walk-in Intake:** Rapid intake form capturing demographic details, emergency contacts, blood type, and insurance information.
- **Token & Queue System:** Issue sequential consultation tokens and route patients dynamically to waiting rooms.
- **Scheduling Desk:** Book appointments for any available physician, with emergency triage override options.

7. Patient Portal Guide

Empowers patients to take charge of their healthcare journey with direct digital access.

- **Self-Service Appointment Booking:** Select departments, view physician profiles, choose date/time slots, and confirm appointments.
- **Medical Records Vault:** View and download past consultation summaries and digital prescriptions.
- **Digital Billing & Payments:** Inspect itemized bills and complete payments online.

8. Pharmacy Portal Guide

Connects doctors directly with the hospital pharmacy for streamlined medication fulfillment.

- **Prescription Queue:** Real-time feed of electronic prescriptions generated during consultations.
- **Dispensing Workflow:** Verify medicine specifications, check availability, and mark orders as *Dispensed*.

9. Shared & AI-Powered Features

- **MediSync AI Assistant:** Intelligent clinical assistant for symptom triage, differential diagnosis hints, and note formatting.
- **Emergency Medical Wallet:** Quick-access emergency profile storing blood group, severe allergy logs, chronic conditions, and emergency contact details for instant triage.
- **Real-Time Notifications:** Toast and header alerts for emergency admissions, scheduling updates, and bill payments.

10. API & Data Specifications

The backend REST API handles system communication with JWT bearer token authorization.

Endpoint	Method	Description
/api/auth/login	POST	User authentication and JWT token generation
/api/patients	GET / POST	Fetch patient list or register a new patient
/api/doctors	GET / POST	Retrieve doctors directory or add a new doctor
/api/appointments	GET / POST	Fetch system appointments or schedule new session
/api/appointments/:id/status	PUT	Update appointment lifecycle status
/api/prescriptions	GET / POST	Retrieve e-prescriptions or create a new e-prescription
/api/invoices	GET / POST	Fetch billing records or generate new invoice

11. Troubleshooting & FAQ

Q: How do emergency admissions bypass standard waiting queues?

A: When registering or booking an appointment, setting the priority to **Emergency** immediately alerts attending doctors and places the patient at the top of the queue.

Q: How are digital signatures secured on e-Prescriptions?

A: Doctor signature URLs/Base64 strings are stored securely in doctor profile records. When an e-prescription is generated, the signature is embedded into the prescription document.